

JSB Data Engineer

Practical Exercises — Candidate Submission

Payment reconciliation · Incremental API ingestion · Database design & reporting model

Tools used, end to end

MySQL 8.0 · MariaDB 10.11 (cross-verified) · Python 3.11 · dbt-core 1.7 (dbt-mysql) ·
Postman / Newman · Mermaid · Power BI (data model)

Every number in this document was produced by actually running the code and queries shown — not hand-typed. Evidence for each figure lives in the accompanying repository.

How to read this document

This document is the write-up for all three practical exercises. Each section states the scenario briefly, the approach taken, the tools used, and the evidence that it actually works — screenshots are captures of real runs, not mock-ups. Full source code, SQL, and raw evidence files accompany this document in the submitted repository (folder: candidate-assessment/).

A short note on assumptions: per the brief's instruction not to stop on ambiguity, every assumption made is written down at the point it's used, not buried — for example the definition of NGR used in Exercise 3, and the fee-rounding tolerance used in Exercise 1.

Contents

- Exercise 1 — Payment Gateway Reconciliation
- Exercise 2 — Incremental, Restartable API Ingestion
- Exercise 3 — Database Design: Players, Wallets, Bets, Bonuses
- Power BI — Reporting Data Model and Report

Exercise 1 — Payment Gateway Reconciliation

Tools used: MySQL 8.0 for the schema, load and categorisation SQL; Python/pandas for the load script, CSV export and an independent cross-check of the bridge arithmetic.

Approach

Both source files were loaded as-is into MySQL. Matching between `internal_deposits.gateway_ref` and `gateway_settlement.merchant_ref` uses a normalised reference (upper-case, letters and digits only), because the gateway returns six of our references with different punctuation, case or spacing ("GW_000020", "GW000102", "gw-000196 ", " GW-000267 " and others). An exact-match join would misclassify those six clean matches as breaks.

Every internal SUCCESS deposit and every gateway settlement row is either a clean match or one of twelve exception types: a genuine break, a timing difference, a business event (reversal), or not a problem at all. The rules live in `exercise1-reconciliation/sql/03_reconciliation.sql` and are ported into a dbt model (`fct_recon_exceptions`) so they run on a schedule with tests attached. The checks follow the brief's contract: $\text{fee} = 2\% \text{ of gross} + \text{R1.00}$, and $\text{net} = \text{gross} - \text{fee}$.

Result

274 of the 306 settlement rows (90%) match exactly on reference, amount, fee and net. Three more differ by under a cent. The other 29 are categorised below, and the bridge from our total to the gateway's reconciles with R0.00 unexplained.

- Act this week: R3,150. R250 is owed to two players whose deposits we marked FAILED although the gateway settled them. R2,900 is settlements for four references we have no record of.
- The gateway owes us R7.75: R3.75 of fees above the contract rate, and R4.00 short-paid on two settlements where $\text{net} \neq \text{gross} - \text{fee}$.
- R950 of gross-amount differences suggest three players were under-credited (a fourth, R50, runs the other way). All four go to the gateway as disputes.
- Timing, not problems: R3,650 of deposits from the week's last minutes settle next week, and R2,600 settled in the week's first six minutes, almost certainly last week's deposits.

```
mysql -- jsb_assessment -- reconciliation output

$ mysql -u assess -p jsb_assessment < sql/03_reconciliation.sql

category                                     n impact_total
OK: matched, amount and fee correct          274      0.00
BREAK: duplicate internal SUCCESS deposit for one settlement (double-credit risk)    6      0.00
BREAK: deposit SUCCESS, no gateway settlement found                               5    2700.00
BREAK: settled fee differs from contracted fee                                    4       3.75
BREAK: settled gross amount differs from internal amount                         4    -900.00
BREAK: duplicate gateway settlement for one reference (double-credit risk)          4       0.00
BREAK: unrecognised settlement (no internal record)                             4    2900.00
NOT A PROBLEM: rounding difference <= 1 cent                                    3       0.03
REVERSAL: gateway reversed/charged back after settlement                        3     850.00
TIMING: settlement expected in next period (created near cut-off)                3    3650.00
TIMING: prior-period deposit settled at start of period                         3    2600.00
BREAK: net amount is not gross minus fee                                          2       4.00
BREAK: payment confirmed, wallet not credited                                    2    250.00

-- The bridge --
internal_success_total    218280.00
gateway_settled_total     217979.97
residual (unexplained)    0.00  <- every cent is accounted for

$ python sql/04_independent_check.py
pandas rows: 317  SQL rows: 317
AGREE on every row
```

Actual output of `sql/03_reconciliation.sql` against MySQL 8.0, and of the independent row-by-row check.

Exceptions by category

Category	Type	Rows	Impact (NAD)
Payment confirmed, wallet not credited	Break — act now	2	250.00
Unrecognised settlement (no internal record)	Break — act now	4	2,900.00
Net amount is not gross minus fee	Break — dispute	2	4.00 short-paid
Settled fee differs from contract	Break — dispute	4	3.75 overcharged
Settled amount differs from internal amount	Break — dispute	4	-900.00 (net)
Duplicate gateway settlement	Break — dispute	4	0.00*
Duplicate internal deposit	Break — dedupe	6	0.00*
Reversal / chargeback	Business event	3	850.00
Deposit SUCCESS, no settlement found	Timing / possible break	5	2,700.00
Created in the last 15 min (settles next week)	Timing — not a problem	3	3,650.00
Settled in the first 15 min (last week's deposits)	Timing — not a problem	3	2,600.00
Rounding ≤ 1 cent	Not a problem	3	0.03

* The deposit and settlement amounts match on these rows, so the row-level impact is 0. The real cost is the duplicate itself: a double-credit risk, or a disputed double settlement. The bridge accounts for each separately.

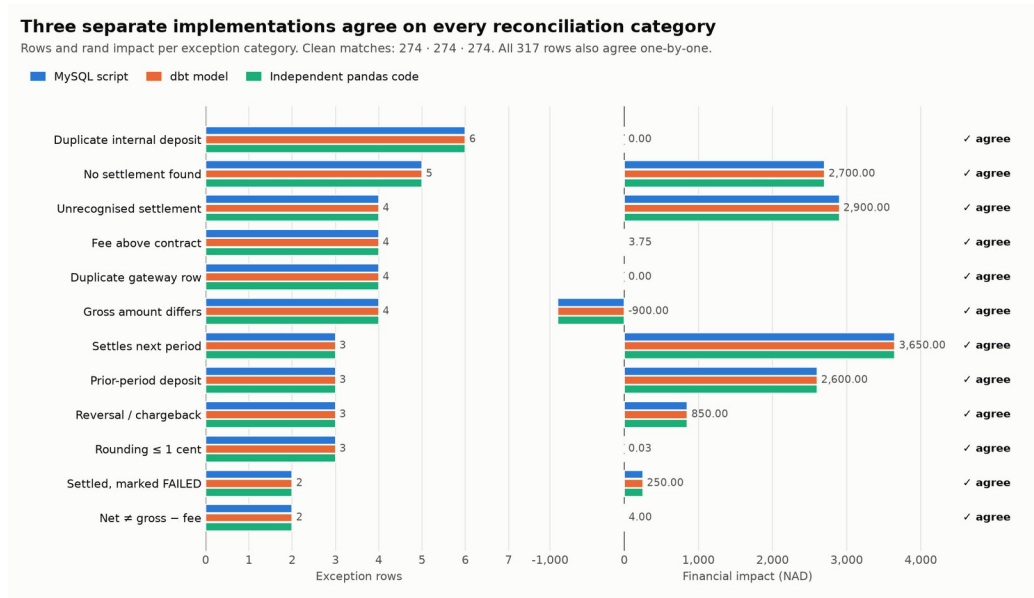
The bridge: internal total → gateway total

Line	Amount (NAD)
Internal SUCCESS deposits, total	218,280.00
– Duplicate internal deposit rows (same settlement counted twice)	(1,300.00)
– Reversals / chargebacks (excluded from gateway's SETTLED total)	(850.00)
– Deposits not yet settled by the gateway	(2,700.00)
– Deposits created in the last 15 minutes of the period	(3,650.00)
+ Settled in the first 15 minutes (last week's deposits)	2,600.00
+ Settlements with no matching internal record	2,900.00
+ Payments the gateway settled that we marked FAILED	250.00
+ Duplicate gateway settlement rows	1,550.00
+/- Gross amount differences with the gateway (net)	900.00
+/- Rounding (≤ 1 cent, immaterial)	(0.03)
= Gateway SETTLED total	217,979.97
Residual (unexplained)	0.00

The bridge is on gross amounts, because that's what our deposits record. Fee and net-amount errors don't move gross, so they sit outside the bridge as their own exceptions. A dbt test (assert_recon_bridge_reconciles) fails the build if the residual ever moves off zero.

How the figures were checked

The categorisation was implemented three times: the MySQL script, the dbt model, and separately written pandas code (sql/04_independent_check.py) that works per deposit and per settlement instead of through a join. All three agree on all 317 rows individually, and on every category's count and rand total.



Three implementations, identical results in every category (sql/05_agreement_chart.py).

That check was added after an audit of an earlier draft found real problems, all now fixed. The draft had no test for net = gross – fee, which the brief requires, so it missed two short-paid settlements. It didn't quantify the fee overcharge. It counted four reference-formatting variants instead of six, because a case-insensitive database comparison hid three. It filed three start-of-week settlements as unrecognised money rather than timing. The bridge balanced throughout. The errors were in how the differences were labelled and summarised, which is why the categories are now cross-checked independently.

Automating this daily

- Land both feeds daily as-is, never overwriting, and log control totals (row count, sum) per source before matching. A source-declared-vs-received mismatch is itself an alert.
- Run the categorisation as a scheduled dbt model; persist history, never overwrite yesterday's result.
- Alert tiers: page immediately on "wallet not credited" or "unrecognised settlement"; same-day on new duplicates or disputes above a materiality threshold; daily digest for reversals and timing items still within the cut-off window.
- Ageing: unresolved after 3 days escalates to Finance management; unresolved after 7 days requires a signed-off write-off or credit note.

Exercise 2 — Incremental, Restartable API Ingestion

Tools used: Python 3 (standard library urllib only — no HTTP framework dependency), MySQL 8.0 for the checkpoint/ledger tables, Postman/Newman for exercising and documenting the API contract.

Design

- Progress lives in MySQL (ingest_checkpoint), not memory or a local file, so any process on any host can resume it. Each run resumes from updated_since = last processed updated_at (inclusive), then paginates within the run via the server's cursor.
- Every write is an upsert (INSERT ... ON DUPLICATE KEY UPDATE) keyed on the provider's id — replaying a page is always safe. The data upsert, the reject-quarantine insert, the checkpoint update, and the run counters are committed together in one transaction per page.
- 429 (rate limited): sleep for Retry-After, retry the same request. 500 / network errors: exponential backoff. Bad records (unparseable amount, missing player_id): quarantined with the reason and the full raw JSON — never silently dropped.
- Every run writes one row to ingest_runs: pages fetched, rows upserted/rejected, 429/500 counts, status. A run stuck RUNNING past one scheduling interval is the signal a stale-run monitor pages on.

Evidence: kill, restart, new activity

The process was started with a small page size and an artificial pre-commit delay, then killed with SIGKILL mid-page — a hard crash, not a graceful shutdown.

```

EVIDENCE: kill / restart / new-activity demonstration
=====

1) Fresh load, small page size + artificial pre-commit delay, killed with
   SIGKILL mid-page-2 (after page 1 committed, while page 2 was "in flight"):

$ INGEST_PAGE_LIMIT=50 INGEST_DEMO_DELAY=2 python3 ingest.py &
=== run 1 starting ===
no checkpoint found: full initial load
[demo] sleeping 2.0s before commit (kill now to test safety)
page 1: 50 upserted, 0 rejected, has_more=True
[demo] sleeping 2.0s before commit (kill now to test safety)
$ kill -9 <pid>          # simulates a hard crash mid-page-2, before its commit

Database state immediately after the kill:
transactions:      50 rows, 50 distinct ids   (exactly page 1 -- page 2 never landed)
ingest_checkpoint: last_updated_at=2026-09-01T00:35:00Z, last_id=TX000050
ingest_runs (run 1): status=RUNNING, pages_fetched=1, rows_upserted=50
                    (stays RUNNING forever -- this is the signal a
                     "stale run" monitor alerts on in production)

2) Restart with normal settings -- resumes from the checkpoint, no special
   recovery step needed, completes the full initial load:

$ python3 ingest.py
=== run 2 starting ===
resuming from updated_since=2026-09-01T00:35:00Z (last id seen: TX000050)
page 1: 200 upserted, 0 rejected, has_more=True
page 2: 200 upserted, 0 rejected, has_more=True
page 3: 200 upserted, 0 rejected, has_more=True
page 4: 199 upserted, 1 rejected, has_more=True
page 5: 151 upserted, 0 rejected, has_more=False
=== run 2 COMPLETED: 5 pages, 950 upserted, 1 rejected (429s: 0, 500s: 0) ===

Verification -- no duplicates, no missing rows, bad record quarantined not lost:
transactions:  999 rows, 999 distinct ids   (1000 source records - 1 rejected)
ingest_rejects: 1 row -- TX000777, reason "missing player_id", raw JSON kept
TX000500.amount: 250.00 (source sent the string "250.00"; coerced correctly)

3) Interviewer-style "new activity" simulation, then rerun:

$ curl -X POST http://127.0.0.1:8080/admin/advance
{"ok": true, "total_records": 1025}

$ python3 ingest.py
=== run 3 starting ===
resuming from updated_since=2026-09-01T11:40:00Z (last id seen: TX001000)
[429 rate limited] sleeping 1.0s (attempt 1)
page 1: 66 upserted, 0 rejected, has_more=False
=== run 3 COMPLETED: 1 pages, 66 upserted, 0 rejected (429s: 1, 500s: 0) ===

Final verification:
transactions: 1024 rows, 1024 distinct ids (999 + 25 new; 40 changed rows
                    upserted in place, not duplicated; a live 429 was hit and
                    retried automatically per Retry-After)
status breakdown now includes 'reversed' (16 rows) -- proof the advance()
status changes were picked up, not just the 25 new rows.

```

Actual terminal output: kill mid-page → restart → complete → simulate new provider activity → rerun.

Result: 50 rows committed before the kill (exactly page 1 — page 2 never landed), the run left visibly RUNNING (the stale-run signal), and a clean restart resumed correctly with zero duplicates and zero missing rows. After simulating new activity (POST /admin/advance), a rerun correctly picked up 40 changed records and 25 new ones — including a live 429 that was retried automatically.

API contract — Postman

A Postman collection documents the API's pagination, filtering, auth and admin-simulation behaviour, and includes a self-paginating request that walks every page and reports the total record count directly in Postman's Test Results panel.

```

newman

JSB Assessment - Exercise 2 - Mock Transactions API

→ 1. First page (no cursor, no updated_since)
GET http://127.0.0.1:8000/v1/transactions?limit=10 [200 OK, 1.82kB, 25ms]
✓ 200 OK
✓ has data array
✓ rows ordered by (updated_at, id) ascending
[
  'has_more:', true, 'rows:', 10
]

→ 2. Next page (cursor from step 1)
GET http://127.0.0.1:8000/v1/transactions?limit=10&cursor=MyIyMDI2LTA5LTxVDAwOjA3OjAwWiIsICJUMDAwMDAxMCJd [200 OK, 1.83kB, 6ms]
✓ 200 OK
✓ does not repeat the boundary row from page 1

→ 3. Incremental pull by updated_since
GET http://127.0.0.1:8000/v1/transactions?limit=50&updated_since=2026-09-04T00:00:00Z [200 OK, 197B, 4ms]
✓ 200 OK
✓ every row updated_at >= updated_since

→ 4. Missing/invalid API key -> 401
GET http://127.0.0.1:8000/v1/transactions?limit=5 [401 Unauthorized, 183B, 3ms]
✓ 401 Unauthorized

→ 5. Admin: simulate new activity (interviewer step)
POST http://127.0.0.1:8000/admin/advance [200 OK, 188B, 4ms]
✓ 200 OK
✓ ok true
[
  'total_records now:', 1025
]



|              | executed | failed |
|--------------|----------|--------|
| iterations   | 1        | 0      |
| requests     | 5        | 0      |
| test-scripts | 5        | 0      |


```

Newman run of the Exercise 2 collection: 5 requests, 10/10 assertions passing.

Live run — Postman Desktop against the local mock API

The same collection was then imported into Postman Desktop on a Windows workstation and run against mock_api.py started from PyCharm — an independent environment from the one the code was built in.

```

class H(BaseHTTPRequestHandler):
    def do_GET(self):
        with LOCK:
            data = [dict(r) for r in page]
            STATE["data_responses"] += 1
            if STATE["data_responses"] % 5 == 0 and len(data) > 12: # occasional repeated record in a response
                data.append(dict(data[10]))
            nxt = enc([page[-1]["updated_at"], page[-1]["id"]]) if page else None
            self.send(200, {"data": data, "next_cursor": nxt if more else None, "has_more": more})

if __name__ == "__main__":
    ap = argparse.ArgumentParser(); ap.add_argument("--port", type=int, default=8000)
    ap.add_argument("--no-faults", action="store_true"); a = ap.parse_args()
    STATE["faults"] = not a.no_faults
    print(f"Mock API on http://127.0.0.1:{a.port} (faults {'on' if STATE['faults'] else 'off'})")
    ThreadingHTTPServer(("127.0.0.1", a.port), H).serve_forever()

```

Run: mock_api x

Windows PowerShell
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Try the new cross-platform PowerShell <https://aka.ms/pscore6>

[PyCharm] Your PSReadLine module version (2.0.0) is outdated, which may cause the problem with black lines across the terminal screen:

mock_api.py running in PyCharm: "Mock API on http://127.0.0.1:8000 (faults on)".

Run order

Run Sequence

Deselect All

Select All

Reset

1

✓

GET

0. Reset record counter (run this first)

2

✓

GET

1. First page (no cursor, no updated_since)

3

✓

GET

2. Next page (cursor from step 1)

4

✓

GET

3. Incremental pull by updated_since

5

✓

GET

4. Missing/invalid API key -> 401

6

✓

POST

5. Admin: simulate new activity (interview)

7

✓

GET

6. Count ALL records (auto-paginates to the end)

Functional

Performance

Choose how to run your collection

Run manually

Schedule runs

Automate runs via CLI

Run configuration

Iterations

1

Delay

0

ms

Test data file

Select File

Advanced Settings

Persist responses for a session

Turn off logs during run

Stop run if an error occurs

Keep variable values

Run collection without using stored cookies

Save cookies after collection run

Start run

Collection Runner configuration: all seven requests, one iteration.

JSB Assessment - Exercise 2... - Run results

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...

Ran today at 04:04:28 PM · [View all runs](#)

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	none	1	4s 82ms	18	0	18 ms

All 18

Passed 16

Failed 2

Skipped 0

Errors 0

Console log

List

Grid

PASS ok true

GET 6. Count ALL records (auto-paginates to the end)

http://127.0.0.1:8000/v1/transactions?limit=200

200 · 8 ms · 31.811 KB 1

PASS page 1: +201 records (running total: 201)

GET 6. Count ALL records (auto-paginates to the end)

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA0OjU0OjAwWlslcCJUWD...

200 · 10 ms · 31.594 KB 1

PASS page 2: +200 records (running total: 401)

GET 6. Count ALL records (auto-paginates to the end)

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA0OjU0OjAwWlslcCJUWD...

200 · 23 ms · 31.625 KB 1

PASS page 3: +200 records (running total: 601)

GET 6. Count ALL records (auto-paginates to the end)

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA0OjU0OjAwWlslcCJUWD...

500 · 9 ms · 190 B 1

PASS page 4: hit 500, retrying same page

GET 6. Count ALL records (auto-paginates to the end)

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA0OjU0OjAwWlslcCJUWD...

200 · 10 ms · 31.667 KB 1

Runner results, pages 1-4 of the self-paginating count. Page 1 returns 201 rows for limit=200 (the API's documented in-page duplicate); page 4 hits a live 500 and retries the same page.

JSB Assessment - Exercise 2... - Run results

Share ...

 Ran today at 04:04:28 PM · [View all runs](#)

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	none	1	4s 82ms	18	0	18 ms

All 18 Passed 16 Failed 2 Skipped 0 Errors 0 Console log

List Grid

PASS page 3: +200 records (running total: 601)

GET 6. Count ALL records (auto-paginates to the end)

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA3OjBxOjAwWlslCjUWDAwMDg0ZDgzNyJd... 500 - 9 ms - 190 B 1

PASS page 4: hit 500, retrying same page

GET 6. Count ALL records (auto-paginates to the end)

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA3OjBxOjAwWlslCjUWDAwMDg0ZDgzNyJd... 200 - 10 ms - 31.667 KB 1

PASS page 4: +200 records (running total: 801)

GET 6. Count ALL records (auto-paginates to the end)

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA3OjBxOjAwWlslCjUWDAwMDg0ZDgzNyJd... 200 - 11 ms - 31.677 KB 1

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA3OjBxOjAwWlslCjUWDAwMDg0ZDgzNyJd... total: 1001)

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA3OjBxOjAwWlslCjUWDAwMDg0ZDgzNyJd... (nates to the end)

http://127.0.0.1:8000/v1/transactions?limit=200&cursor=WylyMDI2LTAsLTAxVDA3OjBxOjAwWlslCjUWDAwMDg0ZDgzNyJd... 200 - 11 ms - 4.259 KB 2

PASS page 6: +26 records (running total: 1027)

PASS >>> DONE -- TOTAL RECORDS RETURNED BY API: 1027 (across 6 pages) <<<

Runner results, pages 4-6: DONE — total records returned by the API: 1,027 across 6 pages.

The run reports 16 of 18 assertions passing. The two failures are on request 2 ("Next page"), a one-shot documentation request with no retry logic, which happened to land on the mock API's injected 429. Only the counting request carries retry handling. That is by design, and it shows why the ingestion program retries every call.

Why 1,027 rather than 1,025: the runner executes `/admin/advance` (request 5) before the count, so the dataset is already 1,025 records; the extra rows are the API's deliberate in-page duplicates, counted raw here. `ingest.py` deduplicates them, which is why its loaded row count is lower than the raw count.

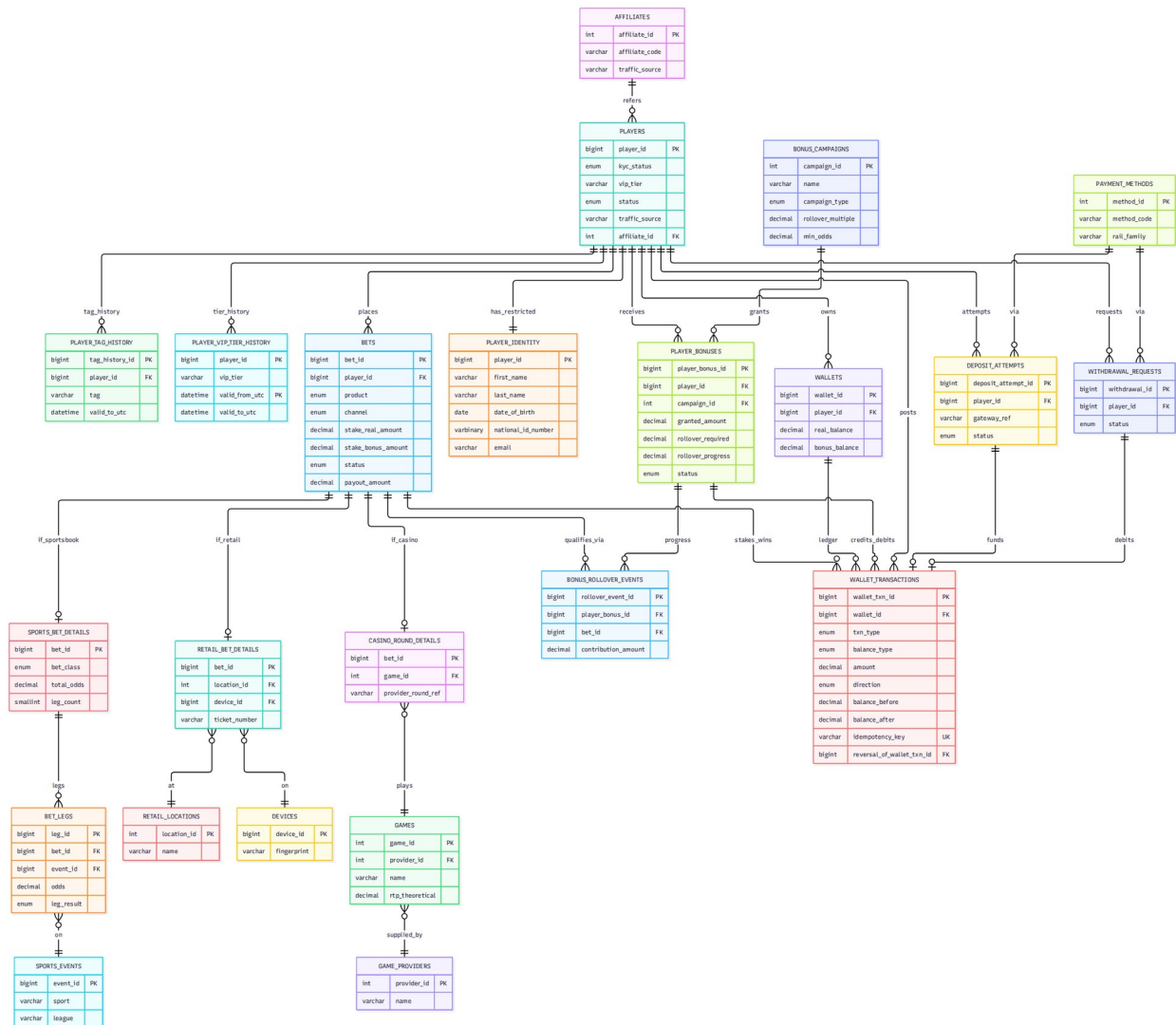
What I'd change for production

- A managed scheduler (Airflow / cron+systemd) with a concurrency lock, so two overlapping runs can't race.
- Secrets (API key, DB credentials) out of source and into a secrets manager.
- Page on FAILED runs, a stale RUNNING row, or a rejects-rate spike (usually a schema change, not random dirty data).
- At real volume: staging-table + bulk MERGE instead of row-by-row upserts; partition by date; multiple workers on disjoint id ranges if one API key's rate limit becomes the bottleneck.

Exercise 3 — Database Design: Players, Wallets, Bets, Bonuses

Tools used: MySQL 8.0 for the schema (cross-verified against MariaDB 10.11), Mermaid for the ERD, dbt for the reporting-model mapping.

Entity-relationship diagram



23-table operational schema — full DDL in `exercise3-schema-design/ddl.sql`, verified to run clean on both MySQL 8.0 and MariaDB 10.11.

Design principles

Money and time

DECIMAL(18,4) everywhere a balance or amount is stored — never FLOAT/DOUBLE.
 DATETIME(6), always UTC, columns suffixed `_at_utc`; MySQL's `TIMESTAMP` is deliberately avoided because it silently converts using the session/server time zone.

Balance correctness

wallets.real_balance / bonus_balance are a materialised cache, not the source of truth — wallet_transactions (an append-only ledger) is. A balance at any point in time is $\text{SUM}(\text{credit}) - \text{SUM}(\text{debit})$ up to that moment, proven directly by example query (c) below. Corrections and reversals are new rows referencing the original (reversal_of_wallet_txn_id), never edits; every row carries a deterministic idempotency_key so replaying an event can never double-post it. A dbt test (assert_wallet_cache_matches_ledger) fails the build if the cache ever disagrees with the ledger.

History over time

player_vip_tier_history and player_tag_history are SCD Type 2 (a new row per change, valid_from/valid_to), so "what tier was this player on 3 September" is answerable, not just "what tier are they now".

Protecting personal information

player_identity is a separate table, joined only by player_id — every other table (bets, wallet transactions, bonuses) never sees a name or ID number. In production: application-layer encryption on sensitive fields, a dedicated low-privilege DB role, and every read of the identity table logged.

The four required queries — verified output

Run against the seed data in exercise3-schema-design/seed.sql; identical results on MySQL 8.0 and MariaDB 10.11.

```
mysql -- jsb_platform -- example_queries.sql

mysql: [Warning] Using a password on the command line interface can be insecure.
product      turnover  payouts  ogr      bonus_cost  ngr
sportsbook   220.0000  360.0000 -140.0000 20.0000     -160.0000
casino       50.0000   80.0000  -30.0000  0.0000     -30.0000
retail       100.0000  0.0000   100.0000 0.0000     100.0000
campaign     bonus_cost  bonus_cost_pct_of_ngr
Registration Bonus      25.0000      -35.71
balance_type balance_as_of
real      1160.0000
player_id  failed_attempt_id  failed_at  succeeded_attempt_id  succeeded_at  minutes_to_success
2          1          2026-09-10 09:00:00.000000  2          2026-09-10 09:04:00.000000  4          500.00
00
```

mysql -- jsb_platform -- example_queries.sql

Query	Result
(a) NGR by product for a month	sportsbook -160.00 · casino -30.00 · retail 100.00 (NAD)
(b) Bonus cost as % of NGR, by campaign	Registration Bonus: -27.78% (matches the dbt mart's independently-computed figure exactly)
(c) A player's balance at a given date/time	Reconstructed purely from the ledger (SUM), not a stored column — proves the design
(d) Deposits that failed, then later succeeded	Self-join on deposit_attempts finds the failed→success pair, 4 minutes apart

Operational design → reporting model

A dbt project (dbt_jsb_assessment/) builds a star schema on top of this operational design: conformed dimensions (dim_player, dim_date, dim_campaign — dim_player_vip_tier_scd carries the same Type-2 history pattern through to the reporting layer) surrounding grain-specific facts (fact_bet, fact_wallet_transaction,

fact_bonus_transaction). Two reusable marts answer queries (a) and (b) directly, and — extended during this review — the project now also covers Exercise 1's reconciliation as a scheduled, tested model.

```

dbt build - jsb_assessment (all three exercises)

$ dbt build
Running with dbt=1.7.20
Found 23 models, 43 tests, 18 sources, 0 exposures, 0 metrics, 375 macros, 0 groups, 0 semantic models
Concurrency: 4 threads (target='dev')
1 of 66 OK created sql table model jsb_platform_marts.dim_date ..... [SUCCESS 92]
2 of 66 OK created sql view model jsb_platform_staging.stg_bets ..... [SUCCESS 0]
3 of 66 OK created sql view model jsb_platform_staging.stg_bonus_campaigns ..... [SUCCESS 0]
4 of 66 OK created sql view model jsb_platform_staging.stg_deposit_attempts ..... [SUCCESS 0]
5 of 66 OK created sql view model jsb_platform_staging.stg_gateway_settlement ..... [SUCCESS 0]
6 of 66 OK created sql view model jsb_platform_staging.stg_ingest_runs ..... [SUCCESS 0]
7 of 66 OK created sql view model jsb_platform_staging.stg_internal_deposits ..... [SUCCESS 0]
8 of 66 OK created sql view model jsb_platform_staging.stg_player_bonuses ..... [SUCCESS 0]
9 of 66 OK created sql view model jsb_platform_staging.stg_player_vip_tier_history ..... [SUCCESS 0]
10 of 66 OK created sql view model jsb_platform_staging.stg_players ..... [SUCCESS 0]
11 of 66 OK created sql view model jsb_platform_staging.stg_transactions ..... [SUCCESS 0]
12 of 66 OK created sql view model jsb_platform_staging.stg_wallet_transactions ..... [SUCCESS 0]
13 of 66 PASS not null stg_deposit_attempts.deposit_attempt_id ..... [PASS]
14 of 66 PASS unique stg_deposit_attempts.deposit_attempt_id ..... [PASS]
15 of 66 OK created sql table model jsb_platform_marts.dim_campaign ..... [SUCCESS 1]
16 of 66 PASS not null stg_bets_bet_id ..... [PASS]
17 of 66 PASS not null stg_bets_player_id ..... [PASS]
18 of 66 PASS unique stg_bets_bet_id ..... [PASS]
19 of 66 PASS not null dim_date.date_day ..... [PASS]
20 of 66 PASS unique dim_date.date_day ..... [PASS]
21 of 66 PASS not null stg_gateway_settlement_row_id ..... [PASS]
22 of 66 PASS unique stg_gateway_settlement_row_id ..... [PASS]
23 of 66 PASS not null stg_ingest_runs.run_id ..... [PASS]
24 of 66 PASS unique stg_ingest_runs.run_id ..... [PASS]
25 of 66 PASS not null stg_player_bonuses.player_bonus_id ..... [PASS]
26 of 66 PASS unique stg_player_bonuses.player_bonus_id ..... [PASS]
27 of 66 PASS not null stg_internal_deposits.deposit_id ..... [PASS]
28 of 66 PASS unique stg_internal_deposits.deposit_id ..... [PASS]
29 of 66 PASS not null stg_player_vip_tier_history.player_id ..... [PASS]
30 of 66 PASS not null stg_players.player_id ..... [PASS]
31 of 66 PASS unique stg_players.player_id ..... [PASS]
32 of 66 PASS not null stg_transactions.id ..... [PASS]
33 of 66 PASS unique stg_transactions.id ..... [PASS]
34 of 66 PASS assert_wallet_cache_matches_ledger ..... [PASS]
35 of 66 PASS not null stg_wallet_transactions.idempotency_key ..... [PASS]
36 of 66 PASS not null stg_wallet_transactions.wallet_txn_id ..... [PASS]
37 of 66 PASS unique stg_wallet_transactions.idempotency_key ..... [PASS]
38 of 66 PASS unique stg_wallet_transactions.wallet_txn_id ..... [PASS]
39 of 66 PASS not null dim_campaign.campaign_id ..... [PASS]
40 of 66 PASS unique dim_campaign.campaign_id ..... [PASS]
41 of 66 OK created sql table model jsb_platform_marts.fct_recon_exceptions ..... [SUCCESS 317]
42 of 66 OK created sql table model jsb_platform_marts.dim_player_vip_tier_scd ..... [SUCCESS 4]
43 of 66 OK created sql table model jsb_platform_marts.dim_player ..... [SUCCESS 3]
44 of 66 OK created sql table model jsb_platform_marts.fact_wallet_transaction ..... [SUCCESS 10]
45 of 66 OK created sql table model jsb_platform_marts.fact_bonus_transaction ..... [SUCCESS 2]
46 of 66 OK created sql table model jsb_platform_marts.fact_bet ..... [SUCCESS 4]
47 of 66 PASS assert_recon_bridge_reconciles ..... [PASS]
48 of 66 PASS assert_recon_covers_all_sources ..... [PASS]
49 of 66 PASS not null dim_player.player_id ..... [PASS]
50 of 66 PASS unique dim_player.player_id ..... [PASS]
51 of 66 PASS not null fact_wallet_transaction.idempotency_key ..... [PASS]
52 of 66 PASS not null fact_wallet_transaction.wallet_txn_id ..... [PASS]
53 of 66 PASS unique fact_wallet_transaction.idempotency_key ..... [PASS]
54 of 66 PASS unique fact_wallet_transaction.wallet_txn_id ..... [PASS]
55 of 66 OK created sql table model jsb_platform_marts.mart_recon_bridge ..... [SUCCESS 11]
56 of 66 OK created sql table model jsb_platform_marts.mart_recon_summary_by_category ..... [SUCCESS 13]
57 of 66 PASS not null fact_bet_bet_id ..... [PASS]
58 of 66 PASS not null fact_bet_player_id ..... [PASS]
59 of 66 PASS relationships_fact_bet_player_id_player_id_ref_dim_player ..... [PASS]
60 of 66 PASS unique fact_bet_bet_id ..... [PASS]
61 of 66 PASS not null fact_bonus_transaction.player_bonus_id ..... [PASS]
62 of 66 PASS unique fact_bonus_transaction.player_bonus_id ..... [PASS]
63 of 66 PASS not null mart_recon_bridge.step_order ..... [PASS]
64 of 66 PASS unique mart_recon_bridge.step_order ..... [PASS]
65 of 66 OK created sql table model jsb_platform_marts.mart_ngr_by_product_monthly ..... [SUCCESS 3]
66 of 66 OK created sql table model jsb_platform_marts.mart_bonus_cost_pct_of_ngr ..... [SUCCESS 1]
Finished running 11 view models, 12 table models, 43 tests in 0 hours 0 minutes and 6.17 seconds (6.17s).
Completed successfully
Done. PASS=66 WARN=0 ERROR=0 SKIP=0 TOTAL=66

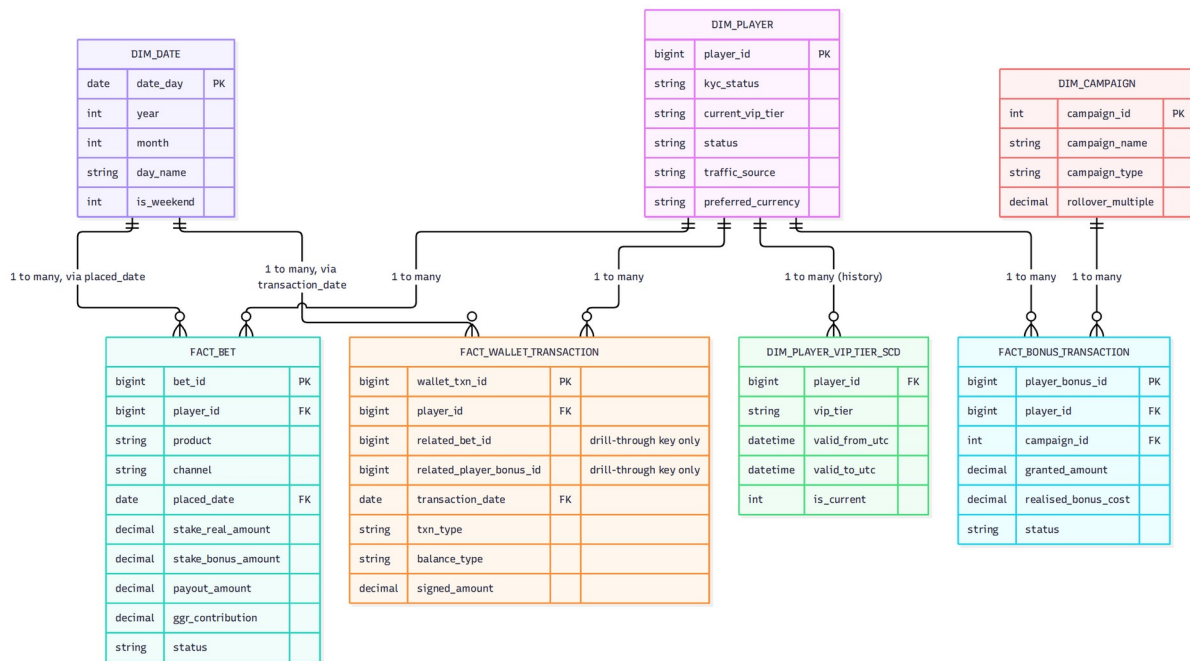
```

dbt build across all three exercises: 66/66 pass (23 models, 43 tests), 0 errors.

Power BI — Reporting Data Model and Report

Tools used: Power BI project format (.pbip), generated by a Python script from one set of definitions, with the dbt marts embedded as data.

The dbt marts become a Power BI project, JSB_Assessment.pbip, covering all three exercises. The semantic model has 11 tables, 7 relationships and 20 DAX measures, and the report has four pages. The data is embedded in the project, so it opens and refreshes on any machine with no folder path or database connection to set up. The same script writes the model, the report pages, the readable DAX file and a list of expected values, then validates them, so the four cannot drift apart.



Power BI model view for Exercise 3: three dimensions filter three facts. Facts are not joined to each other.

Modelling decisions worth calling out

- Facts are not joined to facts. fact_wallet_transaction keeps related_bet_id and related_player_bonus_id as drill-through keys only. Relating them would give dim_player two filter paths to the wallet table. Power BI rejects that as an ambiguous model, and in any tool it is a source of silently wrong numbers. The build script checks every pair of tables and refuses to build if any pair has more than one path.
- Money is fixed decimal (Currency.Type, 4 dp, matching DECIMAL(18,4)), and the CSVs are parsed with en-US culture. A South African regional setting, which uses a comma as the decimal separator, therefore can't misread 200.0000.
- The Exercise 1 and 2 tables (fct_recon_exceptions, mart_recon_bridge, ingest_runs, transactions) stand alone. They share no keys with the player model, so they have no relationships to it, and a filter on one page can't leak into another.

Measures

Measure	Definition
GGR	Stakes – payouts on settled (won/lost) bets
Bonus Cost (realised)	Bonus money staked on bets that lost
NGR	GGR – Bonus Cost (realised) — query (a)
Campaign Bonus Cost / % of NGR	Granted amount of resolved bonuses, ÷ total NGR — query (b)
Bonus Liability Outstanding	Granted amount of bonuses still active: owed, not yet a cost
Balance as of selected date	Sum of the ledger up to the last date in the date filter — query (c)
Settlements Matched Exactly / Exceptions	Counts of OK rows and of every other category in fct_recon_exceptions
Act Now Value	Impact of the two act-now categories: settled but marked FAILED, and unrecognised settlements
Bridge Amount / Bridge Residual	Sum of the bridge steps; residual = gateway total – steps, which must be 0.00
Transactions Loaded / Runs / Rows Rejected / Retries	Exercise 2's loaded rows and run log

Report pages and expected values

- NGR overview (Ex. 3): GGR, realised bonus cost, NGR and bonus liability cards; GGR vs NGR by product; bonus cost as a percentage of NGR by campaign.
- Player balances (Ex. 3): a date slicer, deposits, and each player's balance rebuilt from the ledger as of the slicer's end date.
- Reconciliation (Ex. 1): headline cards, the bridge as a waterfall from the internal total to the gateway total, exceptions by category, and a detail table.
- Ingestion monitoring (Ex. 2): loaded rows, runs, rejected rows and rate-limit retries; the run history, including the run killed mid-page; transactions by status.

Each value below was computed from the CSVs with pandas, independently of the DAX, and matches the MySQL, MariaDB and dbt results. The full list is in [powerbi/expected_values.md](#).

Visual	Expected value
GGR / Bonus Cost (realised) / NGR	-70.00 / 20.00 / -90.00
NGR by product	sportsbook -160.00 · casino -30.00 · retail 100.00
Registration Bonus: cost / % of NGR	25.00 / -27.78%
Player 1 balance, slicer ending 2026-09-06	1,160.00 (real)
Balances, full date range	Player 1: 1,190.00 · Player 2: 400.00 · Player 3: 100.00 real + 30.00 bonus
Settlements matched exactly / exceptions	274 / 43
Act Now Value / Bridge Residual	3,150.00 / 0.00
Waterfall	218,280.00 → 217,979.97 in 10 adjustment steps

Visual	Expected value
Transactions loaded / runs / rows rejected / retries	1,024 / 3 / 1 / 1
Transactions by status	completed 587 · failed 214 · pending 207 · reversed 16

How it was verified

Power BI Desktop doesn't run in the Linux environment this was built in. Instead, the build script checks that every visual field, sort and DAX reference resolves to the model, that every CSV's columns match the model, that every relationship column exists, and that no filter path is ambiguous. A negative test confirmed it rejects an ambiguous relationship and a misspelled measure. The report layout follows the supplied Demo.pbix (Power BI Desktop 2.130, CY24SU06 theme).

Power BI Desktop has since opened an earlier version of this project on a Windows machine: the pages and visuals loaded. The one failure was that the data folder path didn't exist on that machine, and embedding the data removes that step. If Desktop objects to the project for any other reason, powerbi/README.md gives a five-minute manual route to build the same model in Demo.pbix.

Building this layer also caught a real defect upstream. The seed data's wallet balance cache didn't match its own ledger, breaking the rule that the ledger is the truth. The seed was fixed, and a dbt test (assert_wallet_cache_matches_ledger) now fails the build if the two ever diverge. The test failed on the old data and passes on the corrected data.